

Spectral Norms of Independent-Entry Matrices with Regular Moment Growth

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Research draft — 14 September 2026

AI-assisted draft; the upper bound is machine-checked. The upper comparison of Theorem 1.1 has been formalized in Lean 4 and checked by the Lean kernel. The formal target carries no proof gap, and its transitive axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`; Section 11 records the exact statement, the development, and the two places where the formal proof deliberately differs from the informal one. The lower comparison in Section 9 is a known result of Łatała–Świątkowski, reproduced here for completeness, and is *not* formalized. AI tools assisted the derivation, the formalization and the exposition under the author’s direction. There has been no external peer review. The upper comparison is registered with the Palomar registry as entry extttPALOMAR-2026-09-14-000006, whose mechanical verification replayed the development through the Comparator and NanoDa kernels; registration is not acceptance, endorsement, or peer review.

Abstract

We prove the upper comparison in problem MI-32, that is, the upper half of Conjecture 4.3 of Łatała–Świątkowski: for real matrices with independent mean-zero entries whose moments satisfy $\|X_{ij}\|_{L_{2p}} \leq \alpha \|X_{ij}\|_{L_p}$, the expected operator norm is at most C_α times the sum of the row and column variance scales and the shared-index deletion parameter built from weak bilinear moments at order $\log(k+1)$, including the exponent $\log 2 < 1$ at $k = 1$.

The local estimate comes from representing each entry as an independent sign times its magnitude. Sign parity supplies orthogonal sectors while every magnitude power stays inside the polynomial coefficients, so revisiting an entry does not lose its moment information. Creation localizes a few original rows and annihilation a few original columns, and a net on whichever axis is small costs an absolute factor at the relevant moment order. The remaining ingredient is a moment comparison on the cone of polynomials with nonnegative original monomial coefficients, which contains every vacuum walk polynomial and is preserved by the parity projections: in degree d such a polynomial obeys an L_2 -to- L_4 bound with factor $(\sqrt{3}\alpha^2)^d$.

The argument is self-contained. In place of the strong/weak moment comparison for Banach-space-valued sums we prove the independent-coordinate statement the matrix argument actually needs, by an independent copy and a multinomial expansion; in place of the variance-envelope matrix estimate used in the published reduction of the deletion parameter we raise the screening threshold from b^2 to b^3 and estimate the far remainder by an elementary centered Gram second moment. All constants are explicit, and all of them are constants that a proof assistant has checked. We also reproduce the established lower comparison, with attribution, and record precisely what the formal development does and does not certify.

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1 Introduction and the exact mean-norm statement

Let $X = (X_{ij})_{i,j \leq n}$ be a real matrix with independent mean-zero entries. We assume that every absolute moment is finite and that one constant $\alpha \geq 1$ satisfies

$$\|X_{ij}\|_{L_{2p}} \leq \alpha \|X_{ij}\|_{L_p}, \quad p \geq 1, \quad i, j \in [n]. \quad (1)$$

Here $[n] = \{1, \dots, n\}$ and $\|Z\|_{L_p} = (\mathbb{E}|Z|^p)^{1/p}$ for every $p > 0$. At $p < 1$ this notation denotes a moment functional, not a norm. All logarithms below are natural. Matrix norms without a subscript are operator norms for the Euclidean norms on domain and range. For a scalar random variable we also use the abbreviation $\|Z\|_p = \|Z\|_{L_p}$; for a deterministic vector, the subscript 2 denotes its Euclidean norm.

Define the deterministic variance scales

$$\begin{aligned} \sigma_r(X) &= \max_i \left(\sum_j \mathbb{E}X_{ij}^2 \right)^{1/2}, \quad \sigma_c(X) = \max_j \left(\sum_i \mathbb{E}X_{ij}^2 \right)^{1/2}, \\ M(X) &= \sigma_r(X) + \sigma_c(X), \quad \sigma(X) = \max\{\sigma_r(X), \sigma_c(X)\}. \end{aligned} \quad (2)$$

For a deterministic set $I \subset [n]$ and $p > 0$, put

$$R_{X,I}(p) = \sup_{\|s\|_2, \|t\|_2 \leq 1} \left\| \sum_{i,j \notin I} X_{ij} s_i t_j \right\|_{L_p}, \quad R_X(p) = R_{X,\emptyset}(p). \quad (3)$$

The vectors in this supremum are deterministic. The deletion parameter is

$$D(X) = \max_{1 \leq k \leq n} \min_{I \subset [n], |I| \leq k} R_{X,I}(\log(k+1)). \quad (4)$$

One and the same original set I deletes both rows and columns. It is chosen for the law, before any realization is observed. The minimum is outside the vector supremum and is attained, since the collection of sets is finite. Deleting all indices gives zero, so the $k = n$ term is zero. The $k = 1$ term retains its exact exponent $\log 2 < 1$.

Theorem 1.1 (Independent entries with regular moment growth). *For every $\alpha \geq 1$ there are constants $c_\alpha, C_\alpha > 0$ such that every $n \geq 1$ and every matrix satisfying (1) obey*

$$c_\alpha \{M(X) + D(X)\} \leq \mathbb{E}\|X\| \leq C_\alpha \{M(X) + D(X)\}. \quad (5)$$

The constants depend only on α , not on the dimension, the individual entry distributions, or the deterministic deletion sets.

Only the upper comparison in (5) is proved here. It is the new content of this paper, and it is the half that has been machine-checked; an explicit admissible value of C_α is recorded in (76). The lower comparison is a known result, reproduced with attribution in Section 9. Section 1.2 locates both halves in the literature and says precisely what the present argument adds.

1.1 The proof mechanism

For symmetric entries, write $Y_e = \varepsilon_e Z_e$ in law, where $Z_e = |Y_e|$ and the original signs are independent of all magnitudes. The signs give an orthogonal parity basis. Magnitude powers stay inside its coefficients, so visiting an entry repeatedly does not lose its moment information. In the row-to-column direction, parity creation has exact sectors involving few original rows, whereas parity annihilation has different exact sectors involving few original columns. A net on this small axis has constant cost at the relevant moment order.

The remaining estimate is a polynomial moment comparison on the cone of nonnegative original monomial coefficients. That cone contains every vacuum walk polynomial and is preserved by the parity projections and every subsequent original multiplication. It is not the entire linear span of these polynomials. This distinction permits the argument to retain rare-amplitude effects without bounding an unrestricted orthogonal-polynomial multiplication operator.

Every analytic input is proved below from (1): the Boolean fourth-moment inequality (Lemma 2.2), the positive-cone comparison (Lemma 2.4), the scalar and vector moment comparisons of Section 3, and the deletion reduction of Section 8. See Remark 2.1.

1.2 Relation to the literature

Estimate (5) is the content of [5, Conjecture 4.3], recorded as problem MI-32 in [10]. Its lower half is [5, Theorem 4.1], which we reproduce in Section 9; the upper half is what is proved here. The conjecture refines a long line of work on the norm of a matrix with independent but not identically distributed entries: Seginer’s comparison with the largest row and column [13], Latała’s variance-based bound [2], the sharp Gaussian and sub-Gaussian estimates of Bandeira–van Handel [1], the logarithmic-factor results of Riemer–Schütt [12], and the dimension-free structure theory of Latała–van Handel–Youssef [6]; see [14] for a survey. For weighted Rademacher matrices the conjecture is known up to a triple logarithm [3], and for general independent entries up to an iterated logarithm [9]; the present estimate removes the remaining gap under the regularity hypothesis (1).

The relation to [5] is worth stating precisely, because it isolates exactly what is new. [5, Remark 4.5] shows that, to obtain the upper half of their conjecture, it suffices to prove the *undelated* local estimate

$$\mathbb{E}\|X\| \leq C_\alpha \{M(X) + R_X(\log(n+1))\}. \quad (6)$$

Corollary 7.2 below is exactly (6) for entries symmetric in law, and it is the new local estimate of this paper. The deduction of the conjecture from (6) sketched in [5, Remark 4.5] passes through a variance-envelope bound taken from [6, Theorem 4.4]. Section 8 gives a different deduction that avoids that input; the reason and the cost are explained there and in Remark 8.6.

1.3 Organization

Sections 2 and 3 prove the polynomial, scalar and vector moment estimates used later. Section 4 sets up the parity–amplitude representation and Section 5 the two count decompositions. Sections 6 and 7 prove the local moment theorem and its consequence (6) at the exact order $\log(n+1)$. Section 8 replaces the undeleted weak moment by the deletion parameter and removes the symmetry assumption, which completes the upper bound. Section 9 reproduces the known lower comparison, Section 10 delimits what is claimed, and Section 11 describes the machine verification.

2 Polynomial moment estimates from the entry law

Every analytic ingredient of the upper bound is proved in this section and the next, from the entry hypothesis (1) alone. Two features of this are worth stating at the outset, because earlier treatments of the same route depend on external matrix theorems.

Remark 2.1 (No external analytic input). The upper bound uses no strong/weak moment comparison for Banach-space-valued sums, and no variance-envelope matrix estimate. The

vector comparison the argument needs is the independent-coordinate statement of Theorem 3.2, which is proved here by an independent copy and a multinomial expansion. The far remainder of the deletion argument is estimated in Section 8.4 by an elementary centered Gram second moment. Consequently the only results quoted from the literature are the lower comparison of Section 9, which is not part of the estimate proved here, and the classical Boolean fourth-moment inequality of Lemma 2.2, for which we give a complete proof.

2.1 Hilbert-valued Boolean fourth moments

On independent signs $\varepsilon_1, \dots, \varepsilon_N$, write $\varepsilon_S = \prod_{e \in S} \varepsilon_e$ and let N_ρ be the noise operator defined on Fourier characters by $N_\rho \varepsilon_S = \rho^{|S|} \varepsilon_S$. Only the L_2 -to- L_4 case of Boolean hypercontractivity is needed, and only for Hilbert-space-valued coefficients. We give its short proof; the general Boolean theorem is in [11, Ch. 9–10].

Lemma 2.2 (Hilbert-valued Boolean polynomials). *If $F = \sum_{|S| \leq d} \varepsilon_S v_S$ takes values in a real Euclidean space, then*

$$\|F\|_{L_4(\ell_2)} \leq 3^{d/2} \|F\|_{L_2(\ell_2)}. \quad (7)$$

Proof. For fixed real Hilbert vectors a, b ,

$$\mathbb{E}_\varepsilon \|a + \varepsilon b\|_2^4 = (\|a\|_2^2 + \|b\|_2^2)^2 + 4\langle a, b \rangle^2 \leq \|a\|_2^4 + 6\|a\|_2^2 \|b\|_2^2 + \|b\|_2^4.$$

If A, B depend on other signs, integrate this inequality and use Cauchy–Schwarz in the mixed term. Taking a square root gives

$$\|A + \varepsilon B\|_{L_4(\ell_2)}^2 \leq \|A\|_{L_4(\ell_2)}^2 + 3\|B\|_{L_4(\ell_2)}^2. \quad (8)$$

Set $\rho = 1/\sqrt{3}$ and induct on the number of signs. Write $F = A + \varepsilon_N B$, where A, B depend on the first $N - 1$ signs. Denoting by N'_ρ the noise operator on those signs, the induction hypothesis and (8) yield

$$\|N_\rho F\|_{L_4(\ell_2)}^2 \leq \|N'_\rho A\|_{L_4(\ell_2)}^2 + 3\rho^2 \|N'_\rho B\|_{L_4(\ell_2)}^2 \leq \|A\|_{L_2(\ell_2)}^2 + \|B\|_{L_2(\ell_2)}^2 = \|F\|_{L_2(\ell_2)}^2.$$

The base case of zero signs is equality. Finally set $G = \sum_{|S| \leq d} \rho^{-|S|} \varepsilon_S v_S$, so $N_\rho G = F$. Hilbert Fourier orthogonality gives

$$\|F\|_{L_4(\ell_2)} \leq \|G\|_{L_2(\ell_2)} = \left(\sum_{|S| \leq d} \rho^{-2|S|} \|v_S\|_2^2 \right)^{1/2} \leq \rho^{-d} \|F\|_{L_2(\ell_2)}.$$

□

2.2 Product moments of the magnitudes

Fix independent symmetric variables $W = (W_e)_{e \in N}$ satisfying (1), and put $Z_e = |W_e|$. For an exponent vector $\nu \in \mathbb{N}_0^N$, write $Z^\nu = \prod_e Z_e^{\nu_e}$ and $|\nu| = \sum_e \nu_e$. Polynomial degree below always means this ordinary total degree, including all repeated powers.

Lemma 2.3 (A product-moment bound). *For $m_e(u) = \mathbb{E} Z_e^u$, $u \in \mathbb{N}_0$, and $m_e(0) = 1$,*

$$m_e(u + v) \leq \alpha^{u+v} m_e(u) m_e(v) \leq \alpha^{2(u+v)} m_e(u) m_e(v), \quad u, v \in \mathbb{N}_0. \quad (9)$$

Consequently, every scalar polynomial $G(Z) = \sum_\nu a_\nu Z^\nu$ with $a_\nu \geq 0$ and degree at most D satisfies

$$\mathbb{E} G(Z)^2 \leq \alpha^{4D} (\mathbb{E} G(Z))^2. \quad (10)$$

Proof. For positive integers u, v , Cauchy–Schwarz and moment doubling give

$$\begin{aligned} m_e(u+v) &\leq \sqrt{m_e(2u)m_e(2v)}, \\ m_e(2u) &= \|W_e\|_{L_{2u}}^{2u} \leq \alpha^{2u} \|W_e\|_{L_u}^{2u} = \alpha^{2u} m_e(u)^2. \end{aligned}$$

Apply the second formula also with v to prove the first inequality in (9); the second holds because $\alpha \geq 1$. If either exponent is zero, the assertion follows from $\alpha \geq 1$; this does not apply moment doubling at zero. Identically zero variables cause no exception.

For two monomials of G , independence and the weaker form of (9) give

$$\mathbb{E}Z^{\nu+\mu} = \prod_e m_e(\nu_e + \mu_e) \leq \alpha^{2(|\nu|+|\mu|)} \prod_e m_e(\nu_e) \prod_e m_e(\mu_e).$$

Multiply by $a_\nu a_\mu \geq 0$, sum over ν, μ , and use $|\nu| + |\mu| \leq 2D$ to obtain (10). $\square$

Lemma 2.4 (Positive-coefficient polynomial comparison). *Let $F(W)$ be a polynomial taking values in a fixed finite-dimensional real Euclidean space. Suppose every coordinate has nonnegative coefficients in the ordinary monomial basis of the original W_e , and $\deg F \leq d$. With*

$$H_\alpha = \sqrt{3}\alpha^2, \tag{11}$$

one has

$$\|F\|_{L_4(\ell_2)} \leq H_\alpha^d \|F\|_{L_2(\ell_2)}, \tag{12}$$

$$\|F\|_{L_{2p/(p-2)}(\ell_2)} \leq H_\alpha^{4d/p} \|F\|_{L_2(\ell_2)}, \quad p \geq 4. \tag{13}$$

Proof. Represent $W_e = \varepsilon_e Z_e$ in law with independent signs and independent magnitudes. Group ordinary monomials by their exponent parity:

$$F(\varepsilon Z) = \sum_S \varepsilon_S F_S(Z).$$

Every coordinate of each F_S still has nonnegative coefficients. Its monomials have exponent parity S , degree at most d , and hence $|S| \leq d$. Conditional on Z , Lemma 2.2 gives

$$\mathbb{E}_\varepsilon \|F(\varepsilon Z)\|_2^4 \leq 3^{2d} \left(\sum_S \|F_S(Z)\|_2^2 \right)^2. \tag{14}$$

The scalar polynomial $G(Z) = \sum_S \|F_S(Z)\|_2^2$ has nonnegative coefficients and degree at most $2d$. Conditional sign orthogonality gives $\mathbb{E}G = \mathbb{E}\|F\|_2^2$. Thus (10), with $D = 2d$, proves

$$\mathbb{E}\|F\|_2^4 \leq 3^{2d} \alpha^{8d} (\mathbb{E}\|F\|_2^2)^2 = H_\alpha^{4d} (\mathbb{E}\|F\|_2^2)^2.$$

Its fourth root is (12). For $u = 2p/(p-2) \in [2, 4]$, the interpolation identity $1/u = (1-4/p)/2 + (4/p)/4$ proves (13). Degree zero gives equality in both comparisons. $\square$

Remark 2.5 (A sharper cone constant, not used). The first inequality of (9) is stronger than the one used, and yields $\mathbb{E}G^2 \leq \alpha^{2D} (\mathbb{E}G)^2$ in place of (10), hence $H_\alpha = \sqrt{3}\alpha$. We deliberately carry the conservative value $\sqrt{3}\alpha^2$, because it is the one the formal development certifies (Section 11); every constant displayed in this manuscript is therefore a machine-checked constant. Using the sharp value replaces one factor α^2 by α at each occurrence of H_α below, and changes nothing else.

The hypothesis of Lemma 2.4 concerns coefficients, not pointwise positivity of $F(W)$. A coordinate such as $W_1 + W_2$ is allowed although its value may be negative. Nor does the lemma apply to every polynomial merely because the law is symmetric: arbitrary negative monomial coefficients may create cancellations in the L_2 norm. We will only project or multiply polynomials in ways that preserve the stated coefficient cone.

2.3 Scalar linear forms double at every even order

The following corollary is the only consequence of Lemma 2.4 needed for scalar sums. It is what replaces the scalar moment-growth estimate that an external strong/weak comparison would otherwise supply.

Corollary 2.6 (Even-order doubling for linear forms). *Let $W = (W_e)$ be independent symmetric variables satisfying (1), let a_e be arbitrary real coefficients, and put $S = \sum_e a_e W_e$. For every integer $q \geq 1$,*

$$\|S\|_{L_{4q}} \leq H_\alpha \|S\|_{L_{2q}}. \quad (15)$$

Proof. Represent $W_e = \varepsilon_e Z_e$. Replacing a_e by $|a_e|$ changes neither side: the signs $\text{sgn}(a_e)\varepsilon_e$ have the same joint law as the ε_e , and both moments depend only on that law. So assume $a_e \geq 0$. Then S^q is a scalar polynomial in the original variables of degree q whose ordinary coefficients are nonnegative, being multinomial coefficients times products of the a_e . Lemma 2.4, applied to the one-dimensional Euclidean space with $d = q$, gives

$$\mathbb{E}S^{4q} = \mathbb{E}|S^q|^4 \leq H_\alpha^{4q} (\mathbb{E}|S^q|^2)^2 = H_\alpha^{4q} (\mathbb{E}S^{2q})^2.$$

Taking $4q$ -th roots gives (15). $\square$

3 A strong/weak comparison for independent coordinates

The thin-matrix estimate of Section 6 needs to compare the L_{4q} moment of the Euclidean norm of a random vector with its variance and its weak moments. For a vector with *independent* coordinates this can be proved directly, and that is all the matrix argument uses. We do so here. No comparison for general Banach norms or general index sets is required, and none is assumed.

3.1 An even-moment comparison for independent symmetric families

Lemma 3.1 (Coordinatewise even moments tensorize). *Let $(A_j)_{j \in J}$ and $(B_j)_{j \in J}$ be finite families of symmetric real random variables, each family independent, with moments through order $2q$ finite. The two families may live on different probability spaces, or on one space with arbitrary dependence between them. If $c \geq 0$ and*

$$\mathbb{E}A_j^{2r} \leq c^{2r} \mathbb{E}B_j^{2r} \quad (j \in J, 1 \leq r \leq q),$$

then

$$\mathbb{E}\left(\sum_j A_j\right)^{2q} \leq c^{2q} \mathbb{E}\left(\sum_j B_j\right)^{2q}. \quad (16)$$

Proof. Expand the $2q$ -th power by the multinomial formula. Independence factors each monomial expectation as $\prod_j \mathbb{E}A_j^{k_j}$ with $\sum_j k_j = 2q$; every such monomial is integrable, by independence and finiteness of the individual powers. Symmetry makes $\mathbb{E}A_j^{k_j} = 0$ whenever k_j is odd, so only terms with all k_j even survive. For those terms every factor is nonnegative and the hypothesis applies coordinatewise; the multinomial coefficients are nonnegative integers, and the powers of c multiply to $c^{\sum_j k_j} = c^{2q}$. Summing the surviving terms gives (16). Exponent-zero factors equal one by normalization. $\square$

Note what this argument does *not* do: it does not compare expansions of absolute values. The odd monomials vanish exactly, and the remaining ones are compared term by term.

3.2 The comparison

Theorem 3.2 (Independent-coordinate strong/weak comparison). *Let $(V_j)_{j \in J}$ be a finite family of independent real random variables, each symmetric in law, with finite moments of order $4q$ for an integer $q \geq 1$. Suppose $\beta \geq 0$ satisfies*

$$\|V_j\|_{L_{4r}} \leq \beta \|V_j\|_{L_{2r}} \quad (j \in J, 1 \leq r \leq q). \quad (17)$$

Put

$$R = \left(\sum_j V_j^2 \right)^{1/2}, \quad \sigma^2 = \sum_j \mathbb{E} V_j^2, \quad W_s = \sup_{\sum_j t_j^2 \leq 1} \left\| \sum_j t_j V_j \right\|_{L_s}.$$

Then

$$\|R\|_{L_{4q}} \leq \sigma + 2\beta^2 W_{2q}. \quad (18)$$

Proof. All the displayed quantities are finite: $R \leq \sum_j |V_j|$ pointwise, Minkowski controls the latter at every order used, and every unit-ball coefficient satisfies $|t_j| \leq 1$, so $W_s \leq \sum_j \|V_j\|_{L_s} < \infty$. The supremum defining W_s is a supremum of deterministic numbers; no measurable selection is involved. For $J = \emptyset$ every quantity vanishes.

Work on a product space carrying an independent copy (V'_j) of the whole family, and set

$$A_j = V_j^2 - (V'_j)^2, \quad B_j = V_j V'_j.$$

The pairs (V_j, V'_j) are independent in j , so (A_j) and (B_j) are independent families. Each A_j is symmetric by exchanging the two copies, and each B_j is symmetric because the law of V_j is symmetric and the copy is independent. These are statements about laws; no sign-flipping map on the underlying space is needed.

For $1 \leq r \leq q$, Minkowski, (17) and exact factorization of the two independent copies give

$$\|A_j\|_{L_{2r}} \leq 2\|V_j^2\|_{L_{2r}} = 2\|V_j\|_{L_{4r}}^2 \leq 2\beta^2 \|V_j\|_{L_{2r}}^2 = 2\beta^2 \|B_j\|_{L_{2r}}. \quad (19)$$

Raising (19) to the power $2r$ and applying Lemma 3.1 with $c = 2\beta^2$ yields

$$\left\| \sum_j A_j \right\|_{L_{2q}} \leq 2\beta^2 \left\| \sum_j B_j \right\|_{L_{2q}}. \quad (20)$$

Put $S = \sum_j V_j^2$ and $S' = \sum_j (V'_j)^2$, so $\mathbb{E} S' = \sigma^2$. Jensen in the copy coordinate, applied to $z \mapsto |z|^{2q}$ and then integrated in the first coordinate, gives

$$\|S - \sigma^2\|_{L_{2q}} \leq \|S - S'\|_{L_{2q}} = \left\| \sum_j A_j \right\|_{L_{2q}}. \quad (21)$$

For a fixed deterministic vector v' , homogeneity of the definition of W_{2q} gives $\|\sum_j v'_j V_j\|_{L_{2q}} \leq W_{2q} \|v'\|_2$; this is trivial at $v' = 0$ and otherwise follows by normalizing. Raising to the power $2q$, integrating over the independent copy and taking the root,

$$\left\| \sum_j B_j \right\|_{L_{2q}} \leq W_{2q} \|R\|_{L_{2q}}. \quad (22)$$

This is the integration of a deterministic scalar bound; the inner law is always the original law of V , and the coefficient vector is the independent copy.

Let $x = \|R\|_{L_{4q}}$ and $y = \|R\|_{L_{2q}}$, both finite and nonnegative. Since $\|S\|_{L_{2q}} = x^2$, combining Minkowski with (21), (20) and (22), and then $y \leq x$, gives

$$x^2 \leq \sigma^2 + 2\beta^2 W_{2q} y \leq \sigma^2 + 2\beta^2 W_{2q} x.$$

For nonnegative x, σ, a the inequality $x^2 \leq \sigma^2 + ax$ forces $x \leq \sigma + a$: otherwise $x > \sigma + a \geq 0$ would give $x^2 > (\sigma + a)x \geq \sigma^2 + ax$. Taking $a = 2\beta^2 W_{2q}$ proves (18). $\square$

The hypotheses of Theorem 3.2 are finite-order and concern only the individual coordinates. They contain neither the conclusion nor any vector moment estimate, and there are no dimension factors in (18).

3.3 Doubling at every real order above $\log 2$

The deletion argument of Section 8 compares weak moments at the subunit order $\log 2$ with weak moments at large scheduled orders. The next proposition supplies that comparison from Corollary 2.6 alone, with an explicit constant and no restriction to orders above one.

Proposition 3.3 (Uniform real-order doubling). *Let $S = \sum_e a_e W_e$ be as in Corollary 2.6 and put*

$$J_\alpha = H_\alpha^7 = (\sqrt{3}\alpha^2)^7. \quad (23)$$

Then

$$\|S\|_{L_{2p}} \leq J_\alpha \|S\|_{L_p}, \quad p \geq \frac{1}{2}. \quad (24)$$

In particular (24) holds for every $p \geq \log 2$.

Proof. Write $\beta = H_\alpha \geq 1$. Corollary 2.6 at $q = 1$ gives $\|S\|_{L_4} \leq \beta \|S\|_{L_2}$, that is, $\mathbb{E}|S|^4 \leq \beta^4 (\mathbb{E}|S|^2)^2$. Hölder with the interpolation $2 = \frac{4}{7} \cdot \frac{1}{2} + \frac{3}{7} \cdot 4$ gives

$$\mathbb{E}|S|^2 \leq (\mathbb{E}|S|^{1/2})^{4/7} (\mathbb{E}|S|^4)^{3/7},$$

hence $(\mathbb{E}|S|^2)^7 \leq (\mathbb{E}|S|^{1/2})^4 (\mathbb{E}|S|^4)^3 \leq \beta^{12} (\mathbb{E}|S|^{1/2})^4 (\mathbb{E}|S|^2)^6$, so $\mathbb{E}|S|^2 \leq \beta^{12} (\mathbb{E}|S|^{1/2})^4$ and therefore

$$\|S\|_{L_2} \leq \beta^6 \|S\|_{L_{1/2}}. \quad (25)$$

All these integrals are finite, and the argument is unaffected if one of them vanishes.

Let $\frac{1}{2} \leq p \leq 2$. Monotonicity of $p \mapsto \|S\|_{L_p}$ on a probability space, Corollary 2.6 at $q = 1$ and (25) give

$$\|S\|_{L_{2p}} \leq \|S\|_{L_4} \leq \beta \|S\|_{L_2} \leq \beta^7 \|S\|_{L_{1/2}} \leq \beta^7 \|S\|_{L_p}.$$

Let $p > 2$ and put $r = \lfloor p/2 \rfloor \geq 1$, so that $2r \leq p \leq 4r$ and $2p \leq 8r$. Two applications of Corollary 2.6, at $2r$ and at r , give

$$\|S\|_{L_{2p}} \leq \|S\|_{L_{8r}} \leq \beta \|S\|_{L_{4r}} \leq \beta^2 \|S\|_{L_{2r}} \leq \beta^2 \|S\|_{L_p} \leq \beta^7 \|S\|_{L_p}.$$

Finally $\log 2 > \frac{1}{2}$. □

Taking bilinear suprema over deterministic unit vectors on a fixed deleted submatrix, (24) gives

$$R_{X,I}(2p) \leq J_\alpha R_{X,I}(p), \quad p \geq \log 2, \quad (26)$$

for every deterministic I , since each bilinear form $\sum_{i,j \notin I} X_{ij} s_i t_j$ is a deterministic linear combination of the original entries. The set I is unchanged by this operation.

4 The exact parity–amplitude representation

For now let $Y = (Y_{ij})_{i,j \leq n}$ have independent symmetric entries satisfying (1). There is one original label $e = (i, j)$ for each nonzero entry; denote this finite label set by $\mathcal{E}$. Keeping zero entries as zero multipliers would give the same formulas. Row vertices r_i and column vertices c_j belong to disjoint copies of $[n]$; write $\mathcal{V}$ for their union.

Represent the joint law as $Y_e = \varepsilon_e Z_e$. The magnitudes are independent because the original entries are independent. Signs can be chosen independent of all magnitudes even at zero atoms: the product law on (ε, Z) has exactly the required image law for Y . The sign at (i, j) is distinct from that at (j, i) when both labels exist. Both directions of the single edge between r_i and c_j use the same variable Y_{ij} .

4.1 The space and its polynomial domain

Let μ_Z be the joint magnitude law and define

$$\mathcal{H} = L_2(\mu_Z; \ell_2(\mathcal{V} \times 2^\mathcal{E})), \quad \|f\|_{\mathcal{H}}^2 = \mathbb{E}_Z \sum_{v \in \mathcal{V}} \sum_{S \subseteq \mathcal{E}} |f_{v,S}(Z)|^2. \quad (27)$$

Conditional Fourier transform in the signs is the isometry

$$(\mathcal{U}f)_v(\varepsilon, Z) = \sum_{S \subseteq \mathcal{E}} \varepsilon_S f_{v,S}(Z) \quad (28)$$

onto $L_2(\varepsilon, Z; \ell_2(\mathcal{V}))$. This transform preserves the L_2 norm, not general L_p norms. For example, with one sign and unit magnitude, amplitudes $f_\emptyset = f_{\{e\}} = 1$ have coefficient-vector norm $\sqrt{2}$, whereas $\mathcal{U}f = 1 + \varepsilon_e$ has L_4 norm $2^{3/4}$. Every non- L_2 norm used in the contraction below is therefore a norm of the reconstructed random polynomial $\mathcal{U}f$, rather than of its occupation-amplitude vector. All formulas below may be restricted to polynomial coefficients, for which every required moment is finite. No assertion about boundedness of the full multiplication operator on $\mathcal{H}$ is needed.

Define $\mathcal{P}_d^+$ to be the image under $\mathcal{U}^{-1}$ of vector polynomials in the original Y_e of degree at most d whose ordinary coefficients are nonnegative in each current-vertex coordinate. Thus a monomial $a_{v,\nu} Y^\nu$ contributes $a_{v,\nu} Z^\nu$ to the component (v, S) with $S = \{e : \nu_e \text{ is odd}\}$. This parity compatibility is part of the definition. It ensures that $\mathcal{U}f$ is a polynomial in the original variables, so Lemma 2.4 applies.

The parity grade of (v, S) is $|S|$. Let P_k keep exactly grade k . For $f \in \mathcal{P}_d^+$, necessarily $P_k f = 0$ for $k > d$. Projection onto any collection of pairs (v, S) simply discards ordinary monomials, hence preserves $\mathcal{P}_d^+$. Distinct such collections are orthogonal in (27). Magnitude coefficients on different parity states may have common even powers, including powers of edges outside the occupied set S . They are not required to be orthogonal or independent.

4.2 Multiplication, creation, and annihilation

The actual symmetric dilation is

$$H_Y = \begin{pmatrix} 0 & Y \\ Y^\top & 0 \end{pmatrix}. \quad (29)$$

For a scalar magnitude polynomial $h(Z)$ at state (v, S) , conjugating multiplication by H_Y through $\mathcal{U}$ gives

$$\mathcal{T}(h(Z)|v, S) = \sum_{e=vw} Z_e h(Z) |w, S \triangle \{e\}. \quad (30)$$

Here $e = vw$ means the original bipartite entry with endpoints v, w . Define $\mathcal{C}_+$ by keeping terms with $e \notin S$ and $\mathcal{C}_-$ by keeping terms with $e \in S$. Their sum is exactly $\mathcal{T}$, and they change parity grade by $+1$ and -1 .

Both maps multiply the magnitude coefficient by Z_e . On an ordinary monomial Y^ν , either action increases ν_e by one. Therefore

$$\mathcal{C}_+ \mathcal{P}_d^+ \subset \mathcal{P}_{d+1}^+, \quad \mathcal{C}_- \mathcal{P}_d^+ \subset \mathcal{P}_{d+1}^+, \quad \mathcal{T} \mathcal{P}_d^+ \subset \mathcal{P}_{d+1}^+. \quad (31)$$

In particular annihilation does not divide by a magnitude or reset its earlier powers. On Y_e^3 , its action gives Y_e^4 while removing the odd parity flag. This retains the full fourth moment of that same entry.

5 Exact localization by the two original count vectors

For $S \subset \mathcal{E}$, define

$$r_i(S) = |S \cap (\{i\} \times [n])|, \quad c_j(S) = |S \cap ([n] \times \{j\})|.$$

The vectors $r(S), c(S)$ count occupied original parity labels, not total magnitude degrees. Write δ_i for a coordinate vector in the relevant row or column count space.

Lemma 5.1 (Two exact count decompositions). *Consider maps from a current row to a current column, at input parity grade k . Each of the following gives orthogonal partitions of both the input states and the output states of the specified map.*

1. *For creation, label input (r_i, S) by $d = r(S) + \delta_i$ and output (c_j, U) by $d = r(U)$. Put $I = \{i : d_i > 0\}$. Then $|I| \leq k + 1$ and the entire sector is an input/output parity-count compression of multiplication by $Y[I, :]^\top$.*
2. *For annihilation, label input (r_i, S) by $d = c(S)$ and output (c_j, U) by $d = c(U) + \delta_j$. Put $J = \{j : d_j > 0\}$. Then $|J| \leq k$ and the entire sector is an input/output parity-count compression of multiplication by $Y[:, J]^\top$. At $k = 0$ the map is zero.*

The reverse current-side direction transposes the statements. Every magnitude variable remains available inside the coefficients.

Proof. Creation along $e = (i, j) \notin S$ produces $U = S \cup \{(i, j)\}$, so $r(U) = r(S) + \delta_i = d$. For fixed d , every input current row lies in I , and all parity flags on input and output lie in $I \times [n]$. The sum of the coordinates of d is $k + 1$, giving $|I| \leq k + 1$. Multiplication by $Y[I, :]^\top$ therefore includes every actual transition, with exactly its original coefficient.

More explicitly, let P_d select the input label $r(S) + \delta_i = d$ and Q_d select output grade $k + 1$ with $r(U) = d$. The sector map is $Q_d \mathcal{U}^{-1} M_{Y[I, :]}^\top \mathcal{U} P_d$, with the natural current-coordinate inclusions understood. Output grade removes every annihilation term. Conversely every selected creation term satisfies the same original input and output conditions. This proves equality of maps, rather than a comparison of their coefficients.

For annihilation, $e = (i, j) \in S$ produces $U = S \setminus \{(i, j)\}$, so $c(U) + \delta_j = c(S) = d$. Every parity flag lies in $[n] \times J$ and every output current column lies in J , although the input current row may be arbitrary. Now $\sum_j d_j = k$. Select input label $c(S) = d$ and output grade $k - 1$ with $c(U) + \delta_j = d$. The analogous compression of $M_{Y[:, J]}^\top$ retains exactly the annihilation terms. Creation terms have grade $k + 1$ and cannot survive the output projection.

Each input label is a function of (v, S) alone, and so is each output label. Distinct labels thus give disjoint components of (27), hence orthogonal decompositions. The two current-side directions have orthogonal domains and ranges. Different input grades also have different output grades within each of $\mathcal{C}_+$ and $\mathcal{C}_-$. No amplitude restriction was used: even powers outside the selected row or column set remain in $f_{v, S}(Z)$ throughout. $\square$

The distinction between the two labels is necessary. The creator is controlled through a small set of original rows in its output counts; the annihilator is controlled through a small set of original columns in its input counts. We will estimate the latter directly, not infer it by applying an adjoint argument to a cone-restricted norm inequality.

6 Thin matrices and the cone contraction

Lemma 6.1 (A net only on the small axis). *Let $q \geq 1$ be an integer. For a fixed original row set I and a fixed original column set J ,*

$$(\mathbb{E} \|Y[I, :]\|^{4q})^{1/(4q)} \leq 2 \cdot 5^{|I|/(4q)} \{\sigma_r(Y) + 6\alpha^4 R_Y(2q)\}, \quad (32)$$

$$(\mathbb{E}\|Y[:, J]\|^{4q})^{1/(4q)} \leq 2 \cdot 5^{|J|/(4q)} \{\sigma_c(Y) + 6\alpha^4 R_Y(2q)\}. \quad (33)$$

In particular the numerical factor is at most $2 \cdot 5^{1/4}$ as soon as the small axis has at most q indices. Empty submatrices give zero and require no net.

Proof. Fix a deterministic unit vector $s \in \mathbb{R}^I$ and consider the random vector $Y[I, :]^\top s$, whose j -th coordinate is

$$V_j = \sum_{i \in I} s_i Y_{ij}.$$

Distinct coordinates use disjoint sets of original entries, so the family $(V_j)_j$ is independent; each V_j is a deterministic linear combination of independent symmetric variables, hence symmetric in law. We verify the three hypotheses of Theorem 3.2 for this family.

First, independence and zero means give

$$\sigma^2 = \sum_j \mathbb{E} V_j^2 = \sum_{i \in I} s_i^2 \sum_j \mathbb{E} Y_{ij}^2 \leq \sigma_r(Y)^2. \quad (34)$$

Second, Corollary 2.6 applies to each V_j and supplies (17) with $\beta = H_\alpha$. Third, for a deterministic unit vector t ,

$$\sum_j t_j V_j = \sum_{i \in I, j} s_i t_j Y_{ij},$$

which is one of the bilinear forms in the definition (3) of R_Y evaluated at the unit vectors s (extended by zero) and t ; hence $W_{2q} \leq R_Y(2q)$. Theorem 3.2 therefore gives

$$\| \|Y[I, :]^\top s\|_2 \|_{L_{4q}} \leq \sigma_r(Y) + 2H_\alpha^2 R_Y(2q) = \sigma_r(Y) + 6\alpha^4 R_Y(2q), \quad (35)$$

because $2H_\alpha^2 = 2 \cdot 3\alpha^4 = 6\alpha^4$.

Choose a deterministic $1/2$ -net $\mathcal{N}$ of the unit sphere in $\mathbb{R}^I$, with $|\mathcal{N}| \leq 5^{|I|}$; this cardinality follows by taking a maximal $1/2$ -separated set and comparing its disjoint balls of radius $1/4$ with the ball of radius $5/4$. For every realization, $\|Y[I, :]\| \leq 2 \max_{s \in \mathcal{N}} \|Y[I, :]^\top s\|_2$. Raising to the power $4q$ and bounding a maximum by a sum,

$$(\mathbb{E}\|Y[I, :]\|^{4q})^{1/(4q)} \leq 2 \left(\sum_{s \in \mathcal{N}} \mathbb{E} \|Y[I, :]^\top s\|_2^{4q} \right)^{1/(4q)} \leq 2 \cdot 5^{|I|/(4q)} \{\sigma_r(Y) + 6\alpha^4 R_Y(2q)\},$$

which is (32). Transposing the matrix exchanges the roles of rows and columns and proves (33) with the maximum column variance. If $|I| \leq q$ then $5^{|I|/(4q)} \leq 5^{1/4}$. $\square$

The weak moment entering (32) is taken at order $2q$, half the order of the strong moment on the left. This is a feature of Theorem 3.2 and is what makes the whole chain close at the logarithmic order: the local theorem below needs its weak input at $2q$, not at $4q$.

Proposition 6.2 (Contraction on the preserved cone). *For an integer $q \geq 1$, put*

$$B_\alpha = 2 \cdot 5^{1/4} H_\alpha = 2\sqrt{3} \cdot 5^{1/4} \alpha^2. \quad (36)$$

For every $f \in \mathcal{P}_{q-1}^+$,

$$\begin{aligned} \|\mathcal{C}_\pm f\|_{\mathcal{H}} &\leq B_\alpha \{\sigma(Y) + 6\alpha^4 R_Y(2q)\} \|f\|_{\mathcal{H}}, \\ \|\mathcal{T} f\|_{\mathcal{H}} &\leq 2B_\alpha \{\sigma(Y) + 6\alpha^4 R_Y(2q)\} \|f\|_{\mathcal{H}}. \end{aligned} \quad (37)$$

Proof. Set $p = 4q$. First consider row-to-column creation at a fixed parity grade $k \leq q - 1$ and decompose its input as $\sum_d f_d$ using Lemma 5.1. Every f_d lies in $\mathcal{P}_{q-1}^+$, since its projection only discards ordinary monomials. Its sector has $|I| \leq k + 1 \leq q$. Apply $\mathcal{U}$ first. It identifies the multiplier in that sector with $Y[I, :]^\top$, and the transformed output projection is an L_2 contraction. Hölder is then applied on the original joint probability space to $Y[I, :]^\top \mathcal{U} f_d$. Together, Lemma 2.4 and Lemma 6.1 imply

$$\begin{aligned} \|\mathcal{C}_{+,d} f_d\|_{\mathcal{H}} &\leq \|Y[I, :]^\top \mathcal{U} f_d\|_{L_2(\ell_2)} \\ &\leq (\mathbb{E} \|Y[I, :]\|^p)^{1/p} \|\mathcal{U} f_d\|_{L_{2p/(p-2)}(\ell_2)} \\ &\leq 2 \cdot 5^{1/4} \{\sigma_r(Y) + 6\alpha^4 R_Y(2q)\} H_\alpha^{4(q-1)/p} \|f_d\|_{\mathcal{H}} \\ &\leq B_\alpha \{\sigma_r(Y) + 6\alpha^4 R_Y(2q)\} \|f_d\|_{\mathcal{H}}. \end{aligned} \quad (38)$$

In the last step $4(q-1)/p = (q-1)/q \leq 1$ and $H_\alpha \geq 1$. The expectation in Hölder is over the full joint sign and magnitude law. It requires no independence between the matrix multiplier and $\mathcal{U} f_d$, so magnitude powers outside I stay in the state. The $L_{2p/(p-2)}$ factor is explicitly the norm of $\mathcal{U} f_d$. Only its final L_2 norm is replaced by $\|f_d\|_{\mathcal{H}}$ using the isometry; no L_p isometry of occupation amplitudes is used.

For row-to-column annihilation use its column-count sectors instead. They have $|J| \leq k \leq q - 1$, and (33) gives the same bound with $\sigma_c(Y)$. The reverse current-side direction transposes these two statements. Sum the squared estimates over the orthogonal input and output sectors, over their grades, and over current-side directions. The sums of squared input norms equal $\|f\|_{\mathcal{H}}^2$; there is no factor for the number of sectors. This proves the separate bounds for $\mathcal{C}_+$ and $\mathcal{C}_-$, after replacing σ_r and σ_c by their maximum σ . Their triangle inequality gives the stated bound for $\mathcal{T}$. $\square$

There is no random maximum over row or column sets in this argument. Each set is the support of a deterministic count label, and the same unconditional inequality bounds every corresponding squared sector norm before summation. Nor is (37) a bound on the entire polynomial linear span: it is asserted only on the specified cone at the specified total degree.

7 Vacuum walks and the local moment theorem

Theorem 7.1 (Symmetric regular-law local moment bound). *Let Y be an $n \times n$ matrix with independent symmetric entries satisfying (1). With B_α from (36), every integer $q \geq 1$ satisfies*

$$(\mathbb{E} \|Y\|^{2q})^{1/(2q)} \leq (2n)^{1/(2q)} 2B_\alpha \{\sigma(Y) + 6\alpha^4 R_Y(2q)\}. \quad (39)$$

The same bound holds with $\sigma(Y)$ replaced by any B that dominates both the row and the column variance scale, in particular by $M(Y)$.

Proof. For a current vertex $v \in \mathcal{V}$, let Ω_v be the state with coefficient one at $(v, \emptyset)$ and zero elsewhere. It has norm one in $\mathcal{H}$. Under $\mathcal{U}$ it is the constant vector e_v . For every $j \geq 0$,

$$\mathcal{U} \mathcal{T}^j \Omega_v = H_Y^j e_v.$$

Each coordinate on the right is the sum, over original ordered walks of length j between the indicated vertices, of the product of their entry variables. Every coefficient is a nonnegative integer, even when the realized entries are negative. Repeated edges retain their powers in the same monomial. Thus $\mathcal{T}^j \Omega_v \in \mathcal{P}_j^+$.

Apply Proposition 6.2 successively to the prefixes $j = 0, \dots, q - 1$, keeping the common horizon q :

$$\|\mathcal{T}^q \Omega_v\|_{\mathcal{H}} \leq [2B_\alpha \{\sigma(Y) + 6\alpha^4 R_Y(2q)\}]^q. \quad (40)$$

The dilation H_Y is real symmetric, so $\|H_Y\|^q = \|H_Y^q\|$ and, comparing the operator norm of H_Y^q with its Frobenius norm,

$$\|H_Y\|^{2q} \leq \sum_{v \in \mathcal{V}} \|H_Y^q e_v\|_2^2. \quad (41)$$

Taking expectations, using $\|\mathcal{T}^q \Omega_v\|_{\mathcal{H}}^2 = \mathbb{E} \|H_Y^q e_v\|_2^2$ and $|\mathcal{V}| = 2n$,

$$\mathbb{E} \|H_Y\|^{2q} \leq 2n [2B_\alpha \{\sigma(Y) + 6\alpha^4 R_Y(2q)\}]^{2q}.$$

Finally $\|Y\| \leq \|H_Y\|$, because H_Y maps the embedded copy of a column vector to the embedded copy of its image under Y isometrically on each side. Taking the $2q$ -th root gives (39). Only the row and column variance budgets enter Proposition 6.2, so any common bound B for both may replace $\sigma(Y)$. $\square$

The dimension factor counts both sides of the bipartite dilation. One may instead compare the two diagonal blocks of the even power and replace $2n$ by n ; we retain $2n$ because it is the factor the formal development certifies, and at q comparable to $\log n$ the difference is at most $\sqrt{2}$. Every equality above uses the original joint law, and all expectations are finite by the entry moment assumptions. The proof requires neither polynomial completeness in an infinite L_2 space nor a bounded extension of the full multiplication operator.

Corollary 7.2 (The local mean at the exact logarithmic order). *Let Y be as in Theorem 7.1 with $n \geq 1$, and put*

$$L_\alpha = 4e \cdot 5^{1/4} \sqrt{3} \alpha^2 = 2e B_\alpha. \quad (42)$$

Then

$$\mathbb{E} \|Y\| \leq L_\alpha \{M(Y) + 6\alpha^4 J_\alpha^3 R_Y(\log(n+1))\}. \quad (43)$$

Proof. Take $q = \lceil 2 \log(n+1) \rceil \geq 1$. Then $\log(2n) \leq 2 \log(n+1) \leq q$, so $(2n)^{1/(2q)} \leq e^{1/2} \leq e$, and $2q \leq 4 \log(n+1) + 2 \leq 8 \log(n+1)$ because $\log(n+1) \geq \log 2 > \frac{1}{2}$. Three applications of (26), with $I = \emptyset$, therefore give $R_Y(2q) \leq J_\alpha^3 R_Y(\log(n+1))$. Since $\mathbb{E} \|Y\| \leq (\mathbb{E} \|Y\|^{2q})^{1/(2q)}$, the claim follows from (39) with $B = M(Y) \geq \sigma(Y)$. $\square$

The right-hand side of (43) uses the *undeleted* weak moment $R_Y = R_{Y, \emptyset}$. Replacing it by the smaller deletion parameter $D(Y)$ is exactly the remaining matrix problem, and it is the subject of the next section.

8 From the local estimate to shared-index deletion

Corollary 7.2 controls $\mathbb{E} \|Y\|$ by the undeleted weak moment $R_Y(\log(n+1))$, which can be much larger than $D(Y)$. This section closes that gap. The target is the symmetric-law statement

Theorem 8.1 (Symmetric-law deletion bound). *There is a constant C_α^{sym} , depending only on α , such that every $n \geq 1$ and every $n \times n$ matrix Y with independent entries that are symmetric in law and satisfy (1) obey*

$$\mathbb{E} \|Y\| \leq C_\alpha^{\text{sym}} \{M(Y) + D(Y)\}. \quad (44)$$

One may take

$$C_\alpha^{\text{sym}} = \sqrt{31} + 2\sqrt{1 + \sqrt{2} \alpha^2} + 24e \cdot 5^{1/4} \sqrt{3} \alpha^2 + 144e \cdot 5^{1/4} \sqrt{3} \alpha^6 J_\alpha^{10}. \quad (45)$$

Section 8.7 then removes the symmetry assumption. The overall design of the reduction — a nested deterministic selection of indices, a decomposition into near bands and a far remainder, and a maximum over blocks with growing moment orders — follows Latała–Świątkowski [5, Remark 4.5]. Two steps are different here, and both differences are what make the argument self-contained.

1. The published remainder step invokes a variance-envelope estimate that is a consequence of [6, Theorem 4.4]. Adapting that theorem to the present hypotheses would require a one-sided version of a Gaussian comparison whose stated form assumes two-sided polynomial moment growth, which α -regular entries need not have. We instead raise the per-coordinate screening threshold from b^2 to b^3 and estimate the remainder by an elementary centered Gram second moment (Section 8.4). No general fourth-moment matrix inequality is used.
2. The published argument works with a matrix-symmetric dilation, whose reflected entries repeat variables, and afterwards pays for a paired deletion budget. We instead run the whole decomposition on the original rectangular matrix with one shared index family on both axes — which is precisely what the functional D in (4) quantifies over. Every entry family used below is therefore genuinely independent, and no dilation budget arises.

Throughout this section Y is as in Theorem 8.1, and we abbreviate

$$M = M(Y), \quad D = D(Y), \quad v_{ij} = \mathbb{E}Y_{ij}^2.$$

By (2), $\sum_j v_{ij} \leq M^2$ for every i and $\sum_i v_{ij} \leq M^2$ for every j . Regularity at order two gives the fourth-moment budget

$$\mathbb{E}Y_{ij}^4 \leq \alpha^4 v_{ij}^2, \tag{46}$$

since $\|Y_{ij}\|_{L_4} \leq \alpha \|Y_{ij}\|_{L_2}$.

8.1 The deletion schedule

Define, for $k \geq 0$,

$$b_k = 2^{5^k} - 1, \quad \pi_k = \log(b_k + 1) = 5^k \log 2, \quad t_k = b_k^3. \tag{47}$$

So $b_0 = 1$, $\pi_0 = \log 2$, $b_1 = 31$, and $b_{k+1} + 1 = (b_k + 1)^5$. Three elementary facts are used.

Lemma 8.2 (Schedule arithmetic). *For every $k \geq 0$ and every integer $0 \leq a \leq b_k$,*

$$2(a + a t_k) + b_k \leq b_{k+1}, \tag{48}$$

$$2^k \leq b_k, \quad n \leq b_n \quad (n \geq 0), \tag{49}$$

$$5^{a+1} \leq 125 \cdot 5^{(a-2)^+} \quad (a \geq 1). \tag{50}$$

Proof. Write $b = b_k \geq 1$. Expanding $b_{k+1} = (b+1)^5 - 1$ gives $b_{k+1} = b^5 + 5b^4 + 10b^3 + 10b^2 + 5b$, whereas $2(a + ab^3) + b \leq 2(b + b^4) + b = 2b^4 + 3b$; since $b \geq 1$ this is at most $5b^4 + 5b \leq b_{k+1}$, proving (48). For (49), $b_k + 1 = 2^{5^k} \geq 2^{k+1}$ because $5^k \geq k+1$, and $n+1 \leq 2^n \leq 2^{5^n} = b_n + 1$. Finally (50) is an equality for $a \geq 2$ and reads $25 \leq 125$ at $a = 1$. $\square$

8.2 Two-sided screening and the nested family

Fix a deterministic index set S and an integer $t \geq 0$. For a column $j \in S$, at most t rows i can satisfy $(t+1)v_{ij} > M^2$: otherwise $t+1$ of them would already contribute more than M^2 to $\sum_i v_{ij} \leq M^2$. Let $\text{col}_t(S)$ be the union of S with all such rows, over all $j \in S$, and let $\text{row}_t(S)$ be the transposed construction. Then

$$|\text{col}_t(S)| \leq |S|(1+t), \quad |\text{row}_t(S)| \leq |S|(1+t), \quad (51)$$

and, by construction,

$$j \in S, i \notin \text{col}_t(S) \implies (t+1)v_{ij} \leq M^2, \quad (52)$$

with the analogous statement for row_t and v_{ji} .

For an integer budget $k \geq 1$ let I_k be a deterministic minimizer of $I \mapsto R_{Y,I}(\log(k+1))$ over $|I| \leq k$; the minimum is attained because there are finitely many sets, so

$$R_{Y,I_k}(\log(k+1)) = \min_{|I| \leq k} R_{Y,I}(\log(k+1)) \leq D \quad (1 \leq k \leq n), \quad (53)$$

directly from (4). For $k > n$ we may take $I_k = [n]$, and then $R_{Y,I_k}(p) = 0$ for every $p > 0$, so (53) holds for every $k \geq 1$. Now define the nested family

$$S_0 = I_{b_0} = I_1, \quad S_{k+1} = \text{col}_{t_k}(S_k) \cup \text{row}_{t_k}(S_k) \cup I_{b_k}. \quad (54)$$

Lemma 8.3 (The nested family). *The sets (54) are increasing, and for every $k \geq 0$*

$$|S_k| \leq b_k, \quad S_{n+1} = [n]. \quad (55)$$

Moreover, for $r+1 \leq s$,

$$j \in S_r, i \notin S_s \implies v_{ij} \leq \frac{M^2}{b_r^3 + 1}, \quad i \in S_r, j \notin S_s \implies v_{ij} \leq \frac{M^2}{b_r^3 + 1}, \quad (56)$$

and, for every $a \geq 1$,

$$R_{Y,S_{a-1}}(\pi_{(a-2)+}) \leq D. \quad (57)$$

Proof. Monotonicity is clear from $S \subset \text{col}_t(S)$. Since $|S_0| \leq b_0$, (51) and (48) give $|S_{k+1}| \leq 2|S_k|(1+t_k) + b_k \leq b_{k+1}$ inductively. By (49), $b_n \geq n$, so $I_{b_n} = [n]$ may be taken and $S_{n+1} = [n]$.

For (56), note $i \notin S_s$ and $r+1 \leq s$ imply $i \notin S_{r+1}$, hence $i \notin \text{col}_{t_r}(S_r)$; now (52) with $t = t_r = b_r^3$ gives the first implication, and the transposed construction gives the second.

For (57), enlarging a deleted set can only decrease the weak moment, because masking both deterministic test vectors on the larger set is a Euclidean contraction and reproduces exactly the smaller bilinear form. At $a = 1$ we have $S_{a-1} = S_0 = I_1$ and $\pi_{(a-2)+} = \pi_0 = \log 2 = \log(1+1)$, so (53) at $k = 1$ applies. At $a \geq 2$ we have $I_{b_{a-2}} \subset S_{a-1}$ by (54), and $\pi_{a-2} = \log(b_{a-2} + 1)$, so

$$R_{Y,S_{a-1}}(\pi_{a-2}) \leq R_{Y,I_{b_{a-2}}}(\pi_{a-2}) \leq D$$

by (53) at $k = b_{a-2}$. $\square$

Only b_r , and never b_{r+1} , appears in (56) and (57): the minimizer inserted at step k carries the budget b_k , one stage behind the set that contains it. This one-stage shift is what keeps the recorded logarithmic orders comparable to the scheduled ones in Section 8.5.

8.3 Shells and the exact five-mask partition

By (55) every index lies in some S_k , so

$$\ell_i = \min\{r \geq 0 : i \in S_r\} \in \{0, 1, \dots, n+1\} \quad (58)$$

is well defined, and $\{i : \ell_i \leq r\} = S_r$ by monotonicity. Hence

$$|\{i : \ell_i \leq r\}| \leq b_r. \quad (59)$$

Split the entries of Y by the five deterministic masks

$$\begin{aligned} \text{nearLow} : \ell_i \leq 1 \text{ and } \ell_j \leq 1, & \quad \text{farLow} : \ell_j + 1 < \ell_i, \\ \text{nearHigh} : \lfloor \ell_i/2 \rfloor = \lfloor \ell_j/2 \rfloor \geq 1, & \quad \text{farUp} : \ell_i + 1 < \ell_j, \\ \text{nearCross} : \lfloor (\ell_i + 1)/2 \rfloor = \lfloor (\ell_j + 1)/2 \rfloor \text{ and } \ell_i \neq \ell_j. & \end{aligned} \quad (60)$$

Lemma 8.4 (Exact partition). *For all integers $a, b \geq 0$, exactly one of the five conditions in (60) holds at $(\ell_i, \ell_j) = (a, b)$. Consequently Y is the sum of the five corresponding masked matrices, and*

$$\|Y\| \leq \|Y^{\text{nL}}\| + \|Y^{\text{nH}}\| + \|Y^{\text{nC}}\| + \|Y^{\text{fL}}\| + \|Y^{\text{fU}}\| \quad (61)$$

pointwise, where the superscripts name the masks in the order listed.

Proof. The two far conditions are disjoint from each other and from $|a - b| \leq 1$. For $|a - b| \leq 1$: if $\lfloor a/2 \rfloor = \lfloor b/2 \rfloor$ then either both $a, b \leq 1$, which is nearLow, or $\lfloor a/2 \rfloor \geq 1$, which is nearHigh, and these two are disjoint; otherwise $|a - b| = 1$ with a, b in different pairs $\{2m, 2m+1\}$, which forces $\{a, b\} = \{2m+1, 2m+2\}$ and hence $\lfloor (a+1)/2 \rfloor = \lfloor (b+1)/2 \rfloor = m+1$ with $a \neq b$, which is nearCross and excludes $\lfloor a/2 \rfloor = \lfloor b/2 \rfloor$. Conversely nearCross forces $|a - b| = 1$. The triangle inequality gives (61). $\square$

8.4 The far remainder has a summable Gram energy

Proposition 8.5 (Far orientations). *With Y^{fL} and Y^{fU} as in Lemma 8.4,*

$$\mathbb{E}\|Y^{\text{fL}}\| \leq M\sqrt{1 + \sqrt{2}\alpha^2}, \quad \mathbb{E}\|Y^{\text{fU}}\| \leq M\sqrt{1 + \sqrt{2}\alpha^2}. \quad (62)$$

Proof. Write $A = Y^{\text{fL}}$, so $A_{ij} = Y_{ij}\mathbf{1}\{\ell_j + 1 < \ell_i\}$, and put $w_{ij} = \mathbb{E}A_{ij}^2 \leq v_{ij}$. The masks are deterministic, so the entries of A are still independent, centered and symmetric, and (46) still holds for them. Also $\sum_i w_{ij} \leq M^2$ for every j .

Support cap. Suppose $w_{ik} \neq 0$ and $\ell_j \leq \ell_k$. Then $\ell_k + 1 < \ell_i$, so $i \notin S_{\ell_k+1}$, while $j \in S_{\ell_k}$. Applying the first implication of (56) with $r = \ell_k$ and $s = \ell_k + 1$,

$$w_{ij} \leq v_{ij} \leq \frac{M^2}{b_{\ell_k}^3 + 1}. \quad (63)$$

Pair energy. Fix columns j, k and let $r = \max(\ell_j, \ell_k)$. By symmetry of the claim assume $\ell_j \leq \ell_k$, so $r = \ell_k$. Only indices i with $w_{ik} \neq 0$ contribute, and for those (63) applies, whence

$$\sum_i w_{ij}w_{ik} \leq \frac{M^2}{b_r^3 + 1} \sum_i w_{ik} \leq \frac{M^4}{b_r^3 + 1}. \quad (64)$$

Summation. The number of ordered pairs (j, k) with $\max(\ell_j, \ell_k) = r$ is at most $|\{i : \ell_i \leq r\}|^2 \leq b_r^2$ by (59). Since $b_r \geq 1$ we have $b_r^2/(b_r^3 + 1) \leq 1/b_r$, and $b_r \geq 2^r$ by (49). Therefore

$$\sum_j \sum_k \sum_i w_{ij}w_{ik} \leq M^4 \sum_{r \geq 0} \frac{b_r^2}{b_r^3 + 1} \leq M^4 \sum_{r \geq 0} 2^{-r} = 2M^4. \quad (65)$$

Centered Gram. Let $d_j = \sum_i w_{ij}$, let $\Delta = \text{diag}(d_j)$, and put $G = A^\top A - \Delta$. Distinct original entries are independent, and masking preserves this. For $j \neq k$,

$$G_{jk} = \sum_i A_{ij} A_{ik}$$

is a sum over i of independent centered products, so $\mathbb{E}G_{jk} = 0$ and, the cross terms vanishing,

$$\mathbb{E}G_{jk}^2 = \sum_i \mathbb{E}(A_{ij}^2 A_{ik}^2) = \sum_i w_{ij} w_{ik}. \quad (66)$$

For $j = k$, $G_{jj} = \sum_i (A_{ij}^2 - w_{ij})$ is a sum of independent centered variables, so by (46)

$$\mathbb{E}G_{jj}^2 = \sum_i (\mathbb{E}A_{ij}^4 - w_{ij}^2) \leq \alpha^4 \sum_i w_{ij}^2. \quad (67)$$

No independence among the entries of G is asserted or needed. Since $\alpha \geq 1$, the right-hand sides of (66) and (67) are at most α^4 times the corresponding term of the triple sum in (65), so

$$\mathbb{E}\|G\|_F^2 = \sum_j \sum_k \mathbb{E}G_{jk}^2 \leq 2\alpha^4 M^4. \quad (68)$$

From the Gram energy to the norm. Deterministically $\|A\|^2 = \|A^\top A\| \leq \|\Delta\| + \|G\| \leq M^2 + \|G\|_F$, since $\|\Delta\| = \max_j d_j \leq M^2$ and the operator norm is at most the Frobenius norm. Two applications of Cauchy–Schwarz and (68) give

$$\mathbb{E}\|A\| \leq (\mathbb{E}\|A\|^2)^{1/2} \leq (M^2 + \mathbb{E}\|G\|_F)^{1/2} \leq \left(M^2 + (\mathbb{E}\|G\|_F^2)^{1/2}\right)^{1/2} \leq M\sqrt{1 + \sqrt{2}\alpha^2},$$

which is the first bound in (62). All the integrability used is that of the finitely many fourth entry moments. There is no division by M , so $M = 0$ is included.

For Y^{fU} , note that its transpose is the lower-far mask of $Y^\top$ for the same level function, and the operator norm is invariant under transposition. Running the argument for $Y^\top$ uses the second implication of (56) and the row variance budget, and gives the same bound. $\square$

Remark 8.6 (Why the cube). With the threshold $t_r = b_r^2$ the individual fourth moments are still summable, but (65) would contribute $b_r^2/(b_r^2 + 1) \approx 1$ for every shell, and the number of shells is not bounded in terms of α . Recovering the estimate from that route requires a general fourth-moment matrix inequality — exactly the external input we are avoiding. The cube absorbs the off-diagonal Gram pairs directly, at no cost anywhere else: the enlarged selection is charged only in (48), which the schedule (47) already accommodates.

8.5 The near bands

For $a \geq 0$ put $V_a = \{i : \ell_i \in \{a, a+1\}\}$. Then

$$V_a \subset S_{a+1}, \quad |V_a| \leq b_{a+1}, \quad \text{and} \quad V_a \cap S_{a-1} = \emptyset \quad (a \geq 1), \quad (69)$$

the last because every $i \in V_a$ has $\ell_i \geq a$. Each of the two remaining near masks is block diagonal: Y^{nH} vanishes unless $\lfloor \ell_i/2 \rfloor = \lfloor \ell_j/2 \rfloor$, and Y^{nC} vanishes unless $\lfloor (\ell_i + 1)/2 \rfloor = \lfloor (\ell_j + 1)/2 \rfloor$. The block of Y^{nH} at index m is empty for $m = 0$ and equals the full submatrix $Y[V_{2m}, V_{2m}]$ for $m \geq 1$; the block of Y^{nC} at index m is empty for $m = 0$ and, for $m \geq 1$, is the part of $Y[V_{2m-1}, V_{2m-1}]$ supported off the two diagonal shell blocks.

Lemma 8.7 (Block-diagonal norms and two-colour masks). *Let A be a matrix that vanishes whenever $c(i) \neq c(j)$ for a deterministic map c . Then $\|A\| = \max_m \|A[c^{-1}(m), c^{-1}(m)]\|$; in particular there is no factor for the number of blocks. If moreover σ takes two values on $c^{-1}(m)$ and A^{off} agrees with A where $\sigma(i) \neq \sigma(j)$ and vanishes elsewhere, then $\|A^{\text{off}}[c^{-1}(m), c^{-1}(m)]\| \leq \|A[c^{-1}(m), c^{-1}(m)]\|$.*

Proof. The Euclidean space splits orthogonally along the fibres of c , and A preserves that splitting, which gives the first claim. For the second, let $\Delta = \text{diag}(\pm 1)$ be $+1$ on one value of σ and -1 on the other. Then $A^{\text{off}} = \frac{1}{2}(A - \Delta A \Delta)$ entrywise on the block, and $\|\Delta\| = 1$, so $\|A^{\text{off}}\| \leq \frac{1}{2}(\|A\| + \|A\|) = \|A\|$. $\square$

Proposition 8.8 (Blocks at the scheduled order). *For $a \geq 1$ put $q_a = \lceil 2\pi_{a+1} \rceil$ and*

$$L = L_\alpha \{M + 6\alpha^4 J_\alpha^{10} D\}, \quad (70)$$

with L_α from (42). Then

$$(\mathbb{E}\|Y[V_a, V_a]\|^{2q_a})^{1/(2q_a)} \leq L, \quad \text{and} \quad a + 2 \leq 2q_a. \quad (71)$$

Proof. The submatrix $Y[V_a, V_a]$ has independent symmetric entries satisfying (1), and its row and column variance sums are sub-sums of those of Y , hence at most M^2 .

Weak input. Let s, t be deterministic unit vectors supported on V_a . Since $V_a \cap S_{a-1} = \emptyset$ by (69), the bilinear form $\sum_{i,j \in V_a} Y_{ij} s_i t_j$ is one of the forms occurring in $R_{Y, S_{a-1}}$, so $R_{Y[V_a, V_a]}(p) \leq R_{Y, S_{a-1}}(p)$ for every $p > 0$. Next,

$$2q_a \leq 4\pi_{a+1} + 2 \leq 500 \pi_{(a-2)+} + 4 \log 2 \leq 504 \pi_{(a-2)+} \leq 2^{10} \pi_{(a-2)+},$$

where the second step used (50) in the form $4 \cdot 5^{a+1} \leq 500 \cdot 5^{(a-2)+}$ and $2 \leq 4 \log 2$, and the third used $\pi_{(a-2)+} \geq \log 2$. Ten applications of (26) and then (57) give

$$R_{Y[V_a, V_a]}(2q_a) \leq J_\alpha^{10} R_{Y, S_{a-1}}(\pi_{(a-2)+}) \leq J_\alpha^{10} D. \quad (72)$$

Dimension factor. By (69), $|V_a| \leq b_{a+1}$, and $2b \leq (b+1)^2$ for $b \geq 1$ gives

$$\log(2|V_a|) \leq \log(2b_{a+1}) \leq 2 \log(b_{a+1} + 1) = 2\pi_{a+1} \leq q_a \leq 2q_a,$$

so $(2|V_a|)^{1/(2q_a)} \leq e$. Applying Theorem 7.1 to $Y[V_a, V_a]$ at $q = q_a$ with $B = M$, and inserting (72), gives

$$(\mathbb{E}\|Y[V_a, V_a]\|^{2q_a})^{1/(2q_a)} \leq e \cdot 2B_\alpha \{M + 6\alpha^4 J_\alpha^{10} D\} = L,$$

by (42). Finally $q_a \geq 2\pi_{a+1} = 2 \cdot 5^{a+1} \log 2 \geq 5^{a+1} \geq a + 2$, since $\log 2 > \frac{1}{2}$. $\square$

Lemma 8.9 (A maximum of growing moments). *Let $Z_1, \dots, Z_m \geq 0$ and let $p_i \geq i + 2$ be integers with $\|Z_i\|_{L^{p_i}} \leq L$ for a constant $L > 0$. Then $\mathbb{E} \max_i Z_i \leq 3L$, with no dependence on m and no independence assumption.*

Proof. For $z \geq 0$, $t > 0$ and $p \geq 1$ one has $z \leq t + z^p/t^{p-1}$: this is clear if $z \leq t$, and otherwise $z \cdot t^{p-1} \leq z \cdot z^{p-1}$. With $t = 2L$,

$$\max_i Z_i \leq 2L + \sum_{i=1}^m \frac{Z_i^{p_i}}{(2L)^{p_i-1}}.$$

Taking expectations and using $\mathbb{E} Z_i^{p_i} \leq L^{p_i}$ and $L^{p_i}/(2L)^{p_i-1} = L 2^{-(p_i-1)} \leq L 2^{-(i+1)}$ gives $\mathbb{E} \max_i Z_i \leq 2L + L \sum_{i \geq 1} 2^{-(i+1)} \leq 3L$. $\square$

8.6 Proof of Theorem 8.1

If $L = 0$ in (70), replace L by $L + \epsilon$ throughout and let $\epsilon \downarrow 0$ at the end; all the bounds below are continuous in L . So assume $L > 0$.

The mask Y^{nL} is supported on $S_1 \times S_1$ with $|S_1| \leq b_1 = 31$ by (55). Hence

$$\mathbb{E}\|Y^{\text{nL}}\| \leq (\mathbb{E}\|Y^{\text{nL}}\|_F^2)^{1/2} = \left(\sum_{i \in S_1} \sum_{j \in S_1} v_{ij}\right)^{1/2} \leq (|S_1|M^2)^{1/2} \leq \sqrt{31} M. \quad (73)$$

This is the only block for which no deletion budget is available, and it is also the only one whose dimension is bounded by an absolute constant.

For Y^{nH} apply Lemma 8.7 with $c(i) = \lfloor \ell_i/2 \rfloor$ and $Z_m = \|Y^{\text{nH}}[c^{-1}(m), c^{-1}(m)]\|$, so that $\|Y^{\text{nH}}\| = \max_m Z_m$. The block at $m = 0$ vanishes, and for $m \geq 1$ it equals $Y[V_{2m}, V_{2m}]$, to which Proposition 8.8 applies at $a = 2m \geq 1$ with $p_m = 2q_{2m} \geq 2m + 2 \geq m + 2$. Lemma 8.9 therefore gives $\mathbb{E}\|Y^{\text{nH}}\| \leq 3L$.

For Y^{nC} apply Lemma 8.7 with $c(i) = \lfloor (\ell_i + 1)/2 \rfloor$ and $\sigma(i) = \ell_i \bmod 2$. The block at $m = 0$ vanishes, since there $\ell_i = \ell_j = 0$. For $m \geq 1$ the two levels present in $c^{-1}(m)$ are $2m - 1$ and $2m$, which have opposite parities, so the nearCross condition $\ell_i \neq \ell_j$ is exactly $\sigma(i) \neq \sigma(j)$; the second part of Lemma 8.7 bounds that block by $\|Y[V_{2m-1}, V_{2m-1}]\|$, which Proposition 8.8 controls at $a = 2m - 1 \geq 1$ with $p_m = 2q_{2m-1} \geq 2m + 1 \geq m + 2$. Again $\mathbb{E}\|Y^{\text{nC}}\| \leq 3L$.

Adding these to Proposition 8.5 through (61),

$$\mathbb{E}\|Y\| \leq \sqrt{31} M + 6L + 2M\sqrt{1 + \sqrt{2}\alpha^2}. \quad (74)$$

By (70) and (42),

$$6L = 24e \cdot 5^{1/4} \sqrt{3} \alpha^2 M + 144e \cdot 5^{1/4} \sqrt{3} \alpha^6 J_\alpha^{10} D,$$

so the right-hand side of (74) is $c_1 M + c_2 D$ with $c_1, c_2 \geq 0$ summing to C_α^{sym} as in (45). Since $M, D \geq 0$, this is at most $C_\alpha^{\text{sym}}(M + D)$. $\square$

8.7 Symmetrization without changing the small exponent

For general independent centered entries X , let X' be an independent copy and put $Y = X - X'$. For every $p \geq 1$, conditional Jensen and the triangle inequality give

$$\|X_{ij}\|_{L_p} \leq \|Y_{ij}\|_{L_p}, \quad \|Y_{ij}\|_{L_{2p}} \leq 2\alpha \|Y_{ij}\|_{L_p}.$$

Thus Y has independent entries that are symmetric in law and satisfy (1) with 2α ; moreover $\mathbb{E}Y_{ij}^2 = 2\mathbb{E}X_{ij}^2$, so $M(Y) = \sqrt{2}M(X)$, and $\mathbb{E}\|X\| \leq \mathbb{E}\|Y\|$ by Jensen applied to the convex function $\|\cdot\|$ in the copy coordinate.

For each fixed I, s, t , write the bilinear sum as S and its independent copy as S' . If $p \geq 1$ then $\|S - S'\|_{L_p} \leq 2\|S\|_{L_p}$. If $\log 2 \leq p < 1$, subadditivity of $x \mapsto x^p$ gives instead

$$\|S - S'\|_{L_p} \leq 2^{1/p} \|S\|_{L_p} \leq e \|S\|_{L_p},$$

since $2^{1/p} \leq 2^{1/\log 2} = e$. Because $2 \leq e$, the factor e covers both ranges. Taking the same bilinear suprema, then the same deterministic minimum over each budget — the minimizer for X is a feasible set for Y — and then the budget maximum gives $D(Y) \leq eD(X)$. Every budget keeps its exact order $\log(k + 1)$, and the same original set is deleted from both axes throughout.

Applying Theorem 8.1 to Y with parameter 2α and using $\sqrt{2} \leq e$ therefore proves

$$\mathbb{E}\|X\| \leq C_{2\alpha}^{\text{sym}} \{\sqrt{2}M(X) + eD(X)\} \leq e C_{2\alpha}^{\text{sym}} \{M(X) + D(X)\}, \quad (75)$$

which is the upper comparison in Theorem 1.1 with

$$C_\alpha = e C_{2\alpha}^{\text{sym}}. \quad (76)$$

9 The known lower bound on the same matrix

For completeness we reproduce the argument of [5, Theorem 4.1 and Lemma 4.2], with its small-order and independence details. In this section X itself has independent centered entries satisfying (1). There is no symmetrization or comparison to a second input matrix. Nothing in this section is used for the upper bound, and none of it is part of the formalized statement.

9.1 The scalar growth input

Unlike the upper bound, the lower bound genuinely uses a moment-growth estimate from the literature. By [4, Corollary 1.4], applied to the singleton family of test vectors, there is $K_\alpha \geq 1$ such that every deterministic linear combination S of independent centered variables satisfying (1) obeys

$$\|S\|_{L_p} \leq K_\alpha (p/r)^\beta \|S\|_{L_r}, \quad p \geq r \geq 2, \quad \beta = \max\{1/2, \log_2 \alpha\}, \quad (77)$$

and the exponent β is optimal [4, Remark 1.5]. The same estimate appears as [5, (27)]. Note that Proposition 3.3 is *not* a substitute here: it compares $2p$ with p , whereas the sparsification below needs the growth rate in p itself.

9.2 Row and column variances

Fix a row and put $Q = \sum_j X_{ij}^2$, $v = \mathbb{E}Q$. Regularity at order two and Cauchy–Schwarz give

$$\mathbb{E}Q^2 = \sum_{j,l} \mathbb{E}(X_{ij}^2 X_{il}^2) \leq \alpha^4 \sum_{j,l} \mathbb{E}X_{ij}^2 \mathbb{E}X_{il}^2 = \alpha^4 v^2.$$

Paley–Zygmund implies $\mathbb{E}\sqrt{Q} \geq (4\sqrt{2}\alpha^4)^{-1} \sqrt{v}$. Since every row and column Euclidean norm is at most $\|X\|$,

$$M(X) \leq 8\sqrt{2}\alpha^4 \mathbb{E}\|X\|. \quad (78)$$

Rows or columns with zero variance cause no exception.

9.3 A large weak moment has a small original support

For $p \geq 2$, let $R_X^{\text{sp}}(p)$ be the bilinear supremum restricted to vectors whose supports each have cardinality at most $p^{2\beta}$. Then

$$R_X(p) \leq R_X^{\text{sp}}(p) + C_\alpha M(X). \quad (79)$$

Here is a direct proof. For a unit vector z , set $I_p(z) = \{i : |z_i| \geq p^{-\beta}\}$, so that $|I_p(z)| \leq p^{2\beta}$. Split a bilinear sum into the part on $I_p(s) \times I_p(t)$, the part with $j \notin I_p(t)$, and the remaining

part with $i \notin I_p(s)$ and $j \in I_p(t)$. The first part is controlled by $R_X^{\text{sp}}(p)$. For the second, independence and zero means give

$$\left\| \sum_{i, j \notin I_p(t)} s_i t_j X_{ij} \right\|_2^2 \leq p^{-2\beta} \sum_i s_i^2 \sum_j \mathbb{E} X_{ij}^2 \leq p^{-2\beta} \sigma_r(X)^2.$$

Apply (77) with lower order two to bound its L_p moment by $C_\alpha \sigma_r(X)$. The third part is bounded in the same way by $C_\alpha \sigma_c(X)$. The triangle inequality proves (79), also on every fixed deleted submatrix with the same constant. The union of the two sparse vector supports uses at most $2p^{2\beta}$ original indices.

9.4 A uniform scalar lower tail

Choose $D_\alpha \geq e$ so large that $\|S\|_{L_{2r}} \leq D_\alpha \|S\|_{L_r}$ for $r \geq 2$, using (77), and set $\delta = (4 \log D_\alpha)^{-1} \leq 1/4$. For $P \geq 2/\delta$, take $r = \delta P \geq 2$. Paley–Zygmund applied to $|S|^r$ and scalar moment comparison give

$$\begin{aligned} \mathbb{P}\{|S| \geq \tfrac{1}{2} \|S\|_{L_r}\} &\geq (1 - 2^{-r})^2 \left(\frac{\|S\|_{L_r}}{\|S\|_{L_{2r}}} \right)^{2r} \geq \tfrac{1}{4} e^{-P/2}, \\ \|S\|_{L_r} &\geq K_\alpha^{-1} \delta^\beta \|S\|_{L_P}. \end{aligned}$$

Thus, with $c_0 = \delta^\beta / (2K_\alpha) > 0$,

$$\mathbb{P}\{|S| \geq c_0 \|S\|_{L_P}\} \geq \tfrac{1}{4} e^{-P/2}. \quad (80)$$

Only the scalar comparison of the original independent entries is used.

9.5 Disjoint witnesses for each deletion budget

Fix $1 \leq k \leq n$ and write

$$P = \log(k+1), \quad \gamma_k = \min_{|I| \leq k} R_{X,I}(P).$$

We prove $\gamma_k \leq C_\alpha \mathbb{E} \|X\|$ uniformly in k . Choose $k_0(\alpha)$ large enough that, for $k \geq k_0$,

$$P \geq 2/\delta, \quad 2P^{2\beta} \leq \sqrt{k}, \quad k \geq 4.$$

For $k < k_0$, moment monotonicity and (77) show

$$\gamma_k \leq R_X(P) \leq C_\alpha \max_{ij} \|X_{ij}\|_2 \leq C_\alpha M(X).$$

For $P < 2$, only monotonicity to order two is used, so this covers $k = 1$ with its unchanged exponent $\log 2$. Equation (78) completes these finitely many orders.

Now let $k \geq k_0$. If $\gamma_k \leq 2C_\alpha M(X)$ with the constant from (79), the variance bound again suffices. Otherwise, for every deterministic $|I| \leq k$, sparsification in $X[I^c, I^c]$ supplies unit vectors s, t supported outside I on a common set J of at most $2P^{2\beta} \leq \sqrt{k}$ indices, with

$$\left\| \sum_{i, j \in J} s_i t_j X_{ij} \right\|_{L_P} \geq \gamma_k/2.$$

The suprema are attained: there are finitely many possible supports, and the moment functional is continuous on each compact product of unit balls. Alternatively an arbitrarily small slack gives the same constants.

Apply this observation successively, deleting the union of the previously selected sets. It constructs $m = \lfloor \sqrt{k} \rfloor$ disjoint deterministic sets $J_1, \dots, J_m$ and bilinear sums $S_1, \dots, S_m$ with $\|S_l\|_{L_P} \geq \gamma_k/2$. At every stage the already deleted union has at most $(m-1)\sqrt{k} \leq k$ elements. The random variables S_l are independent because they use disjoint collections of original entries. Each satisfies $|S_l| \leq \|X\|$ pointwise, and (80) gives

$$\mathbb{P}\{|S_l| \geq c_0\gamma_k/2\} \geq \frac{1}{4\sqrt{k+1}}.$$

Consequently

$$\begin{aligned} \mathbb{E}\|X\| &\geq \frac{c_0\gamma_k}{2} \left[1 - \left(1 - \frac{1}{4\sqrt{k+1}} \right)^m \right] \\ &\geq \frac{c_0\gamma_k}{2} \left(1 - e^{-1/(8\sqrt{2})} \right). \end{aligned}$$

In the last step we used $m \geq \sqrt{k}/2$ and $k \geq 4$. Taking the maximum over the original budgets proves

$$D(X) \leq C_\alpha \mathbb{E}\|X\|. \quad (81)$$

Combining (78) and (81) gives the known same-matrix lower estimate. Together with (75), it proves Theorem 1.1. For $n = 1$, $D(X) = 0$ and the direct comparison $\|X_{11}\|_2 \leq \alpha \mathbb{E}|X_{11}|$ also verifies the lower endpoint. All deletion minima throughout this proof remove the same deterministic original set from both axes.

10 Scope and the information retained by the proof

The estimate concerns the expected norm of the same original matrix throughout. Its constants depend only on the entry regularity parameter. The deletion sets are deterministic for the law, use the same original indices on both axes, and retain the exact moment order $\log(k+1)$, including $\log 2 < 1$ at $k = 1$.

Three features of the local argument prevent unsupported reductions. First, the magnitudes and their repeated powers are never replaced by static entry heights or fresh copies. Second, sector orthogonality comes from the original signs and current vertices, not from magnitude monomials on disjoint rows. Third, the multiplier estimate is used only on a cone preserved by the actual walk prefixes; it does not bound all signed linear combinations in a polynomial span.

What is *not* claimed. The constants are explicit but deliberately conservative, and no attempt is made to optimize them; in particular Remark 2.5 already identifies one available improvement that we do not take. No efficient algorithm for evaluating $D(X)$ or $R_X(p)$ is claimed. The moment estimates supply upper-tail bounds through Markov's inequality, but do not by themselves identify matching quantiles, and matching moments are not used to claim matching quantiles. Arbitrary noncommuting coefficient insertions can destroy the nonnegative monomial cone, so applications with such insertions need their own preserved-coefficient argument. Finally, the lower comparison of Section 9 is quoted from [5] and is not proved here in any new way.

11 The formal verification

The upper comparison has been formalized in Lean 4 [7] on top of `mathlib` [8]. This section records what that does and does not establish.

11.1 The formal statement

The formalization fixes the quantities of Section 1 as ordinary definitions over an arbitrary probability space: the moment functional $\|\cdot\|_{L_p}$ as a real power of a Bochner integral, meaningful for $p < 1$; the operator norm as the norm of the Euclidean linear map; M and D exactly as in (2) and (4), with one deterministic set deleted from both axes and the order $\log(k+1)$ unchanged. The compared statement is

```
theorem main_upper (α : ℝ) (hα : 1 ≤ α) :
  ∃ C : ℝ, 0 < C ∧ UpperBoundAt α C
```

where `UpperBoundAt` quantifies over every dimension $n \geq 1$, every measurable space, every probability measure on it, and every matrix of independent centered entries with all absolute moments finite satisfying (1) at every real order $r \geq 1$. No finite-law substitute, conditional comparison, or assumed contraction occurs anywhere, and no estimate is introduced as a hypothesis or as a custom axiom. The transitive axiom audit of the target reports exactly `propext`, `Classical.choice` and `Quot.sound`.

A separate check confirms that the hypothesis class is not vacuous: an independent fair-sign matrix satisfies the hypotheses at the extreme parameter $\alpha = 1$ and has $M = \sqrt{n} > 0$, so (5) is a genuine constraint rather than a statement about an empty class.

11.2 Correspondence with this manuscript

Statement here	Formal name in the MI32 namespace
Theorem 1.1, upper half	<code>main_upper</code>
Lemma 2.2	<code>hilbert_walsh_fourth_le</code>
Lemma 2.3	<code>PositivePolynomial.tensorMoment_add_le</code>
Lemma 2.4	<code>SymmetricPositiveCone.fourth_norm_moment</code>
	<code>SymmetricConeInterpolation.moment_holder_order_le</code>
Corollary 2.6	<code>PositiveLinearPowers.signed_linear_even_root_doubling</code>
Lemma 3.1	<code>SymmetricMomentComparison.sum_even_moment_le</code>
Theorem 3.2	<code>HilbertComparison.independent_coordinate_moment_le</code>
Proposition 3.3	<code>SymmetricLinearAllOrders.linear_root_doubling</code>
Lemma 6.1	<code>ThinMatrix.transposeOperator_moment_four_mul_le</code>
Proposition 6.2	<code>RawCreatorContraction.creator_l2_le</code>
	<code>RawAnnihilatorContraction.annihilator_l2_le</code>
Theorem 7.1	<code>LocalSymmetricMoment.operator_moment_le</code>
Corollary 7.2	<code>LocalLogMoment.mean_spectralNorm_le_literal</code>
Lemma 8.2	<code>DeletionScheduleArithmetic</code>
Lemma 8.3	<code>NestedDeletionFamily</code>
Lemma 8.4	<code>DeletionMasks.entry_sum</code>
Proposition 8.5	<code>FarMaskMean.mean_farLow_le, mean_farUp_le</code>
Lemma 8.7	<code>FiberBlockNorm.norm_le_pi_blockSub</code>
	<code>FiberBlockNorm.norm_offDiagonal_le</code>
Proposition 8.8	<code>NearBlockMoment.block_moment_le</code>
Lemma 8.9	<code>BlockMomentMaximum.mean_max_le</code>
Theorem 8.1	<code>SymmetricDeletion.symmetricUpperBoundAt</code>
Section 8.7	<code>SymmetrizationReduction.upperBoundAt_of_symmetric</code>

11.3 Differences between the two proofs, and limits of the check

The formal proof follows this manuscript, with two deliberate differences, both already flagged where they occur.

1. The cone constant. The formalization retains an earlier, conservative form of the product-moment step, so the certified cone comparison carries $H_\alpha = \sqrt{3}\alpha^2$ rather than the $\sqrt{3}\alpha$ that Lemma 2.3 in fact gives (Remark 2.5). We have quoted the certified value everywhere, so that every displayed constant in this manuscript is machine-checked; the sharp value would improve each of them.
2. The dimension factor in Theorem 7.1 counts both sides of the bipartite dilation, giving $2n$ where the trace comparison between the two diagonal blocks gives n . At the orders used this costs at most $\sqrt{2}$.

The verification is a check of the deduction, not of the modelling. It certifies that the Lean statement follows from the axioms of Lean’s type theory; whether that statement faithfully expresses problem MI-32 is a question about the definitions, which a reader must judge from the Challenge file and from Section 1. The lower comparison of Section 9 is not formalized, so Theorem 1.1 is machine-checked only in its upper half. Independent re-verification by the Comparator and NanoDa tools of the Palomar protocol, an editorial review, and registration are separate steps. All three have now been carried out: commit exttt762bd5e of the development repository is registered as Palomar entry extttPALOMAR-2026-09-14-000006, version 1, whose mechanical report records Comparator, NanoDa and a sandboxed kernel replay admitting only the three standard axioms, and whose editorial review returned no blocking problem. That review is automated; it is not external human peer review, and registration asserts neither novelty nor endorsement.

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